

Appendix R

Detailed Thematic Analysis of the Comparative Evaluation of LitTraceAI

R.0.1 Transparency and Citation Integrity

Transparency Features reflects how experts valued the visibility and inspectability of the relationship between generated content and its supporting evidence. Experts emphasised that LitTraceAI allowed them to trace claims back to their origins, understand how statements were generated, assess the precision of citations, and organize information according to meaningful rhetorical categories.

Eleven out of 28 experts highlighted the importance of directly accessing the original evidence behind generated sentences. One expert explained:

Well, I think the possibility of verifying the source is very good. It is one of the best features of this application [LitTraceAI], because now even ChatGPT recommends papers. But in this model [LitTraceAI], you can directly access the origin of the claim, which makes the process much faster and more efficient. (Expert 4)

Another expert compared LitTraceAI with existing academic AI tools:

SciSpace and ScopusAI are very good, but yours [LitTraceAI] is better. Because it shows in detail where the elements you [LitTraceAI] include come from, the provenance. I find it [provenance] interesting because when you click, it [provenance] explains exactly where it comes from: from which paragraph ... The others only show the citation. (Expert 5)

These accounts suggest that experts valued transparency not only because LitTraceAI cited sources, but because it exposed the evidential pathway underlying the generated content. The ability to inspect the original sentence appeared to increase confidence in the reliability of the output.

This perception was reinforced by other experts who described provenance as a distinctive feature of LitTraceAI compared with existing AI tools. One expert stated, “*Honestly, provenance is a unique feature of this tool*” (Expert 15). Similarly, another experts explained, “*For example, with SciSpace I have no visibility into what it drew on to produce that analysis. The transparency of provenance seems particularly strong and is a distinctive feature of your tool [LitTraceAI].*” (Expert 25). These comparisons suggest that experts perceived transparency not simply as an additional functionality, but as a key differentiator that provided greater visibility into how the content was created.

Another expert emphasised the importance of being able to inspect both the origin and the construction process of generated sentences:

I liked the provenance feature because, if I do not agree with a particular point, I want to see how the text originated. It gives me guidance about where the tool [LitTraceAI] obtained the information and how it generated it. This is something I have not seen in the other applications I use, and I think it is extremely important to know where the sources come from. (Expert 1)

This reflection further illustrates how transparency features enabled experts to critically evaluate the generated content rather than simply accepting the output at face value.

This concern with transparency also extended to understanding how generated sentences were constructed. Nine of the 28 experts emphasised the importance of visibility into the sentence generation process. One expert described the value of “*control and monitoring ... over the generation of each sentence*” and explained that LitTraceAI “*shows you how it does it and is transparent for the researcher*” (Expert 13). Similarly, another expert stated that unlike other tools, this tool “*knows how the text is generated ... and tells you the reason*” (Expert 5). These accounts indicate that explainability was perceived as supporting researcher oversight rather than replacing scholarly judgment.

Another expert highlighted that the tool’s transparency allowed him to inspect both how each sentence was created and the source from which the information was extracted. However, the expert also introduced a more cautious perspective regarding the validity of the explanation process:

The fact that it [LitTraceAI] shows you how each sentence is created and where it gets the information from. Of course, I think this would require another study to analyse in detail whether the justification is correct. (Expert 12)

This comment suggests that, although experts appreciated the visibility of the generation process, some still questioned whether the explanations themselves were fully reliable. Transparency therefore appeared to increase opportunities for verification, while simultaneously making certain experts more aware of the need to evaluate LitTraceAI’s reasoning critically.

Experts also valued mechanisms that allowed them to assess evidential strength. 4 out of 28 experts referred to indicators describing the degree of support behind generated sentences. As one expert explained, “*it gives me the precision of the citation*” (Expert 23). This suggests that experts were actively evaluating the reliability of the generated review rather than passively consuming AI-produced content.

Another expert compared the provenance functionality with similar mechanisms occasionally observed in existing AI systems such as Copilot or ChatGPT:

I have seen this provenance idea sometimes in Copilot, or maybe in ChatGPT, although not with that term. Normally it only appears as a number indicating the source, but it does not break it down the way you [LitTraceAI] do: into different stages, limitations, contrasts, and comparisons. That level of detail offers more means for the researcher to understand where the information comes from and how precise the program’s response is. (Expert 7)

This account suggests that experts valued not only source visibility, but also the granularity of the explanation. Detailed provenance information appeared to support experts in evaluating the precision and trustworthiness of AI-generated content.

At the same time, one expert introduced a broader concern regarding the future reliability of provenance systems in contexts where AI-generated content itself becomes part of the source material. The expert explained, *“I would trust the tool [LitTraceAI] today because the source was written by a person, but I would not trust it if the source information is written by LLMs”* (Expert 3). This suggests that experts’ trust in provenance was still tied to assumptions about the human authorship and authenticity of the underlying literature.

This perspective introduces the concept of **Human-Authorship Trust**, which refers to the extent to which confidence in AI-generated outputs depends on the belief that the underlying source material was created by human authors rather than generated by AI systems. From this viewpoint, trust in provenance is not derived solely from the ability to trace information back to its source, but also from assumptions regarding the authenticity and human origin of the underlying evidence. Consequently, provenance mechanisms may be perceived as more trustworthy when they are linked to human-produced literature, whereas confidence may diminish when source materials are believed to originate from LLM-generated content.

Finally, four experts highlighted the usefulness of categorising information according to their rhetorical functions such as gaps, limitations, and results. One expert explained:

It [LitTraceAI] classifies the source, and this supports the coherence of the content. Having that characterisation works as a consistency guide for structuring the review. The fact that it provides the gaps, limitations, and findings gives a fundamental basis from which to begin constructing a theoretical framework.
(Expert 25)

Another expert noted that the classification functionality allowed him to “*filter gaps and limitations and quickly get an idea of the topic*” (Expert 6). Together, these observations suggest that classification mechanisms were perceived as both organizational and analytical aids that supported literature review development.

Citation Accuracy reflects how experts valued the generation of citations that were both verifiable and accurately connected to the provided set of papers. All experts coded within this sub-theme (5 out of 5) referred to the importance of receiving trustworthy references when evaluating the usefulness of the tool.

Expert 17 described previous negative experiences with general purpose AI systems:

ChatGPT gives me ghost authors. For example, if it gives me five authors, three are real, one is made up, and another exists but belongs to a different paper. In contrast, this tool [LitTraceAI] gives me real citations that I can verify in the bibliography, and I also know they are correct because I am very familiar with the literature on this topic.

This comparison suggests that experts evaluated LitTraceAI based not only on text generation quality, but also on whether the cited references could be traced reliably back to the set of papers. Similarly, Expert 10 noted that “*the added value of this tool is that it provides the correct citations based on the list of papers I sent you.*” This statement indicates further that experts perceived accurate attribution as a central requirement for adopting AI-assisted support in academic writing.

This emphasis on traceability was also connected to broader expectations regarding scientific reliability. Expert 27 explained:

I don't expect it [LitTraceAI] to hallucinate, it is getting data from sources that I provided you, and the provenance means that it can be checked. Those things are really important for the scientific space.

This perspective suggests that experts associated source provenance and verifiability with essential norms of scientific practice. The ability to trace generated sentences back to the set of papers appeared to strengthen confidence in the reliability of the generated content.

R.0.2 Scholarly Synthesis and Academic Discourse

Logical Structure captures experts' perceptions of how LitTraceAI organizes and structures literature review sections through thematic grouping, heading generation, and coherent sequencing of ideas. Seven out of nine experts emphasised the importance of thematic grouping and heading generation. Experts commonly associated these features with the conventional structure of literature review sections. Expert 8 noted that:

How it [LitTraceAI] groups the topics and develops the paragraphs within those topics is very similar to how a related work or background section is written in a paper.

Experts also contrasted LitTraceAI with other AI tools that tend to generate shorter and less structured synthesis. For example, Expert 17 explained:

Another thing is how it [LitTraceAI] groups the topics. This is very similar to the structure of a related work section, which uses headings. Other AIs summarize too much, but this one follows the writing structure of a related work section. Honestly, I like it !!!

These observations suggest that experts valued the tool not only for synthesising literature, but also for reproducing recognizable scholarly writing structures. The use of headings and thematic segmentation appeared to support perceptions of academic appropriateness and readability.

One expert additionally emphasised the practical value of automatic topic organization when working with large collections of papers. Expert 25 explained:

I have not encountered automatic topic-based organisation for a set of papers before. Other tools require clustering instructions to be specified in the prompt. The challenge lies in how to classify papers within a literature set: even when an expert can retrieve them quickly, organising them remains a time-consuming task.

This perspective indicates that experts may perceive automatic thematic organization not only as a writing support feature, but also as a mechanism for reducing the manual effort associated with structuring large literature collections. Moreover, this finding aligns with the concept of **Efficiency Gains** identified in Section 7.1.2. In particular, Expert 3 argued that although a researcher may be capable of identifying comparable topics and structures, doing so would require substantially more time and effort. Accordingly, the value of automatic thematic organisation extends beyond the quality of the generated structure to the efficiency with which large volumes of literature can be analysed, synthesised, and organised.

Further conversations with some experts suggested that they were also impressed by the accuracy and appropriateness of the generated headings. This was reflected in questions such as “*Regarding the UPB organisational identification and the theoretical frameworks and measures: did you create those yourself? And how did you prompt the Artificial Intelligence to do it?*” (Expert 24) and “*I have just one question: who defined the three topics? Was it the tool itself?*” (Expert 22). These reactions may indicate that some experts perceived the

proposed headings as sufficiently meaningful and academically coherent to resemble human guided literature organization.

In addition, two out of nine experts focused on the organisation and coherence of information within paragraphs. One expert stated:

The tool [LitTraceAI] demonstrates a strong capacity to organise information within paragraphs, maintaining a logical and coherent structure that aids comprehension. Each paragraph presents a clearly defined central idea, supported by relevant evidence and connected fluently to the rest of the text.

Together, these findings indicate that experts perceived logical structure both at the macro level through thematic organization and headings, and at the paragraph level through coherent development of ideas.

Critical Analysis describes how experts perceived LitTraceAI as supporting scholarly interpretation through the comparison, contrast, and analytical connection of studies. Experts frequently associated this capability with deeper literature synthesis and the identification of relationships across papers.

6 out of 7 experts highlighted LitTraceAI's ability to contrast findings between studies. Expert 2 explained that *"unlike ScopusAI, which gives me more summarised information, here I see that this AI is more argumentative"* and that LitTraceAI provides *"contrast between studies"*. Similarly, Expert 21 noted that *"the connections, ideas, and contrasts are correct"*, suggesting that the analytical linking of findings contributed to perceptions of rigour.

Several experts also positioned this capability as a distinguishing feature compared to other AI tools. Expert 19 stated:

What I want to highlight about this tool [LitTraceAI], and what clearly sets it apart from others like Consensus or ChatGPT, is its ability to contrast findings. I have not seen that feature in any other tool. When I ask Consensus directly

about gaps in the literature, for instance, it sidesteps the question entirely and returns something altogether different.

Similarly, Expert 7 stated:

However, my experience with other tools is that they tend to oversimplify ideas, removing or overlooking information and elements that could be useful for identifying existing limitations or gaps. This tool [LitTraceAI] seems to focus more on content. It devotes greater reasoning to the analysis of the content itself than to the use of formulaic language typical of information synthesis processes. In other words, it aims to produce a genuine state of the art: it compares ideas, extracts them, and tries to preserve the spirit of scientific writing.

This may indicate that experts valued LitTraceAI not only for summarising literature, but also for supporting forms of synthesis associated with scholarly reasoning. At the same time, Expert 7's comment indicates that this value was partly defined in contrast to a perceived limitation of other tools: the risk of excessive simplification during synthesis.

Additionally, although only 1 expert referred explicitly to inferential interpretation enabled by the organisation of the generated content, this perspective remains analytically relevant. The expert's research background is grounded in qualitative analysis, where generating interpretive premises and identifying implicit meanings are central analytical practices. Expert 1 explained:

Furthermore, while no explicit acknowledgements of limitation are made, the structure of the text allows me to identify constraints such as insufficient funding and limited impact, which the authors do not directly state but that I can infer..

This suggests that, at least for some experts, the structured presentation of findings may facilitate interpretive reasoning beyond explicit textual statements. This observation highlights a form of **Support for Inferential Reasoning**, which refers to the ability of

organised information to facilitate the identification or reconstruction of relationships that are implied rather than explicitly stated in the source material. However, this differs from the concept of **Limitations in Inferential Reasoning** described in Section 7.1.2, which refers to difficulties in identifying or reconstructing relationships that are implied rather than explicitly stated in the source material. Whereas the limitation concerns the tool's inability to make certain implicit conceptual associations, the present finding suggests that the structure generated by LitTraceAI can support researchers in making such inferences themselves.

Narrative Tone refers to how experts perceived the generated literature review section as demonstrating a writing style comparable to human-authored academic discourse. All experts associated with this sub-theme (6 out of 6 experts) described the generated content as natural, academically appropriate, and aligned with scholarly writing conventions. Several experts emphasised that the generated text resembled the writing style typically expected from researchers. Expert 11 stated that:

In contrast, it [the generated content] is as if it was genuinely humanised writing, meaning the content appears to have been written by a researcher with small improvements, but very realistic.

This perception suggests that experts valued the generated content not only for its coherence, but also for its resemblance to authentic academic communication. Similarly, Expert 20 connected this perceived authenticity to the way the tool reproduced the structure and rhetorical style commonly associated with literature reviews:

For me, there is no point of comparison. What makes this tool [LitTraceAI] so sophisticated is that it incorporates a human component: it reflects the logic behind how a state-of-the-art section is built in a research paper, and it replicates that process quite well.

This account may indicate that experts evaluated the generated text according to broader scholarly expectations regarding how literature reviews should be constructed and communicated.

In addition, Expert 12 noted that LitTraceAI “*uses language similar to that used by academics and the content is what I expected*”, reinforcing the perception that stylistic familiarity contributed positively to the evaluation of the generated output.

Overall, these findings indicate that the perceived usefulness of the generated literature review was partly associated with its ability to approximate a credible and recognisable academic voice.

Logical Flow describes how experts perceived the generated literature review section as maintaining coherent and connected arguments across paragraphs. The expert associated this quality with the smooth progression of ideas and the clear relationship between evidence and central claims. Regarding the cohesion and continuity of the generated content, 1 out of 1 expert highlighted the logical organisation of ideas throughout the text. Expert 9 stated:

The tool [LitTraceAI] demonstrates an excellent ability to organize information within paragraphs, maintaining a logical and coherent structure that facilitates understanding of the content. Each paragraph presents a well defined central idea, supported with relevant evidence and connected fluently with the rest of the text.

This may indicate that experts valued not only the presence of relevant information, but also the rhetorical coherence through which the literature was synthesised and presented. Although this perspective was expressed by a single expert, the comment remains analytically relevant given the expert’s engineering background, where academic writing tends to prioritise concise and highly condensed communication. As Expert 9 explained:

Generally, the engineering field is more concrete and straight to the point, without unnecessary elaboration. The aim is to be as concise as possible: stating the

problem, the context, the related studies, and the objective of the paper, in order to show how the problem will be addressed. As a result, the introduction is sometimes only one page long, or two pages at most in rare cases.

This suggests that, in this context, maintaining a smooth progression of ideas may be particularly important for synthesising and condensing relevant information efficiently.

R.0.3 Usability

Usability captures how LitTraceAI facilitates efficient interaction through a simple and accessible interface. One out of one experts emphasised the importance of minimizing interaction complexity and reducing unnecessary navigation steps when accessing generated content. This observation is particularly relevant because Expert 5 specialises in accessibility and usability, providing a more informed perspective on how interface design and navigation influence user interaction with AI systems. As Expert 5 explained:

Because it [LitTraceAI] is easy to use, it does not complicate things: you just click once and the content appears, without having to navigate through multiple options.

The relevance of this insight became more evident when the expert contrasted LitTraceAI with other AI-based literature summarisation tools she regularly uses. Expert 5 explained that “the AIs [she] uses to synthesise literature are not so easy to use, meaning the navigation of the content is very complex.” Referring to SciSpace, she noted that it contains “too many options that are complementary to literature synthesis,” which makes it difficult for users to identify the actual summarisation functionality until they become familiar with the interface. She further explained that “the same happens with Elicit,” observing that while Elicit was initially simple to use, “now it is so overloaded with options that one gets lost.”

Similarly, the expert described usability challenges when interacting with Consensus:

For each sentence, it provides too many citations, and when you click to access a paper, you get lost between the main page flow and the paper content. In some cases, you are even redirected to the official paper link, which breaks the initial navigation experience entirely.

From the perspective of an accessibility and usability expert, these observations may indicate that interface simplicity is not merely a matter of convenience, but an important factor supporting sustained attention, orientation, and confidence during interaction. Expert 5 explained that *“if the navigation flow is complex, the user gets distracted easily and loses interest on the topic.”* She additionally observed that novice users may incorrectly blame themselves when navigation becomes confusing, stating that *“this causes users to think that they did something wrong in the system and that is why they lost the main page.”*

The expert also highlighted the instability introduced by constantly evolving interfaces, explaining that *“one challenge with these AIs is that they constantly change their interfaces, which I infer is intended to benefit the user, but it creates the opposite effect.”* Given Expert 5’s background in accessibility and usability, these reflections may suggest that maintaining navigational simplicity and interface consistency could play an important role in supporting effective engagement with AI-assisted literature synthesis tools.